

Scalable Learning Systems



Acceleration of LLMs

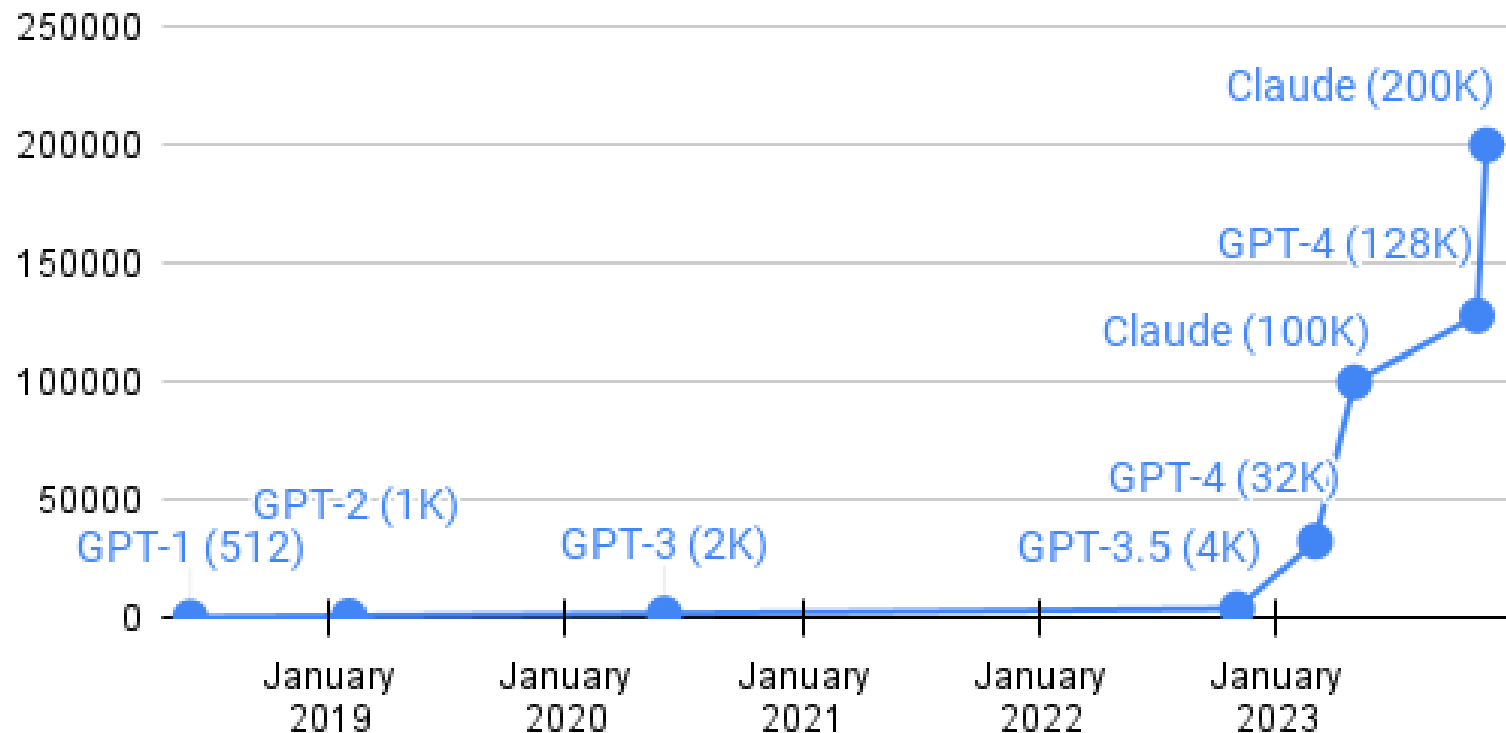
Kubilay Atasu

Flash Attention

From Chris Ré's NeurIPS 2023 Keynote

More (longer) context in the models!

Foundation Model Context Length



Modern GPU Memory Hierarchy

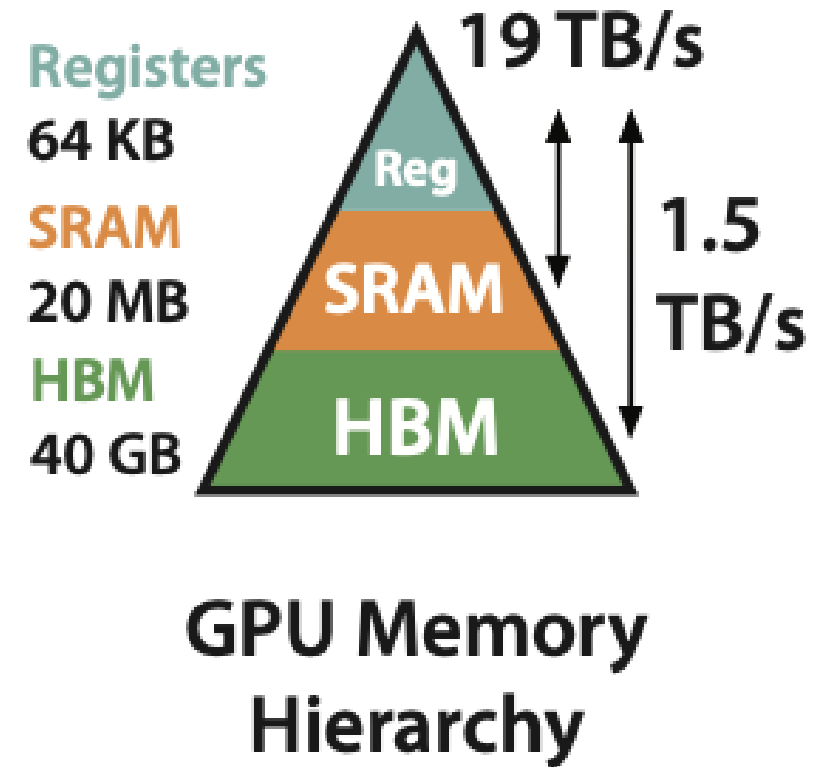


(Simplified) Memory hierarchy

- Registers
- SRAM
- Memory

Small, Fast memory (Registers/SRAM)
Big, Slow memory (HBM)

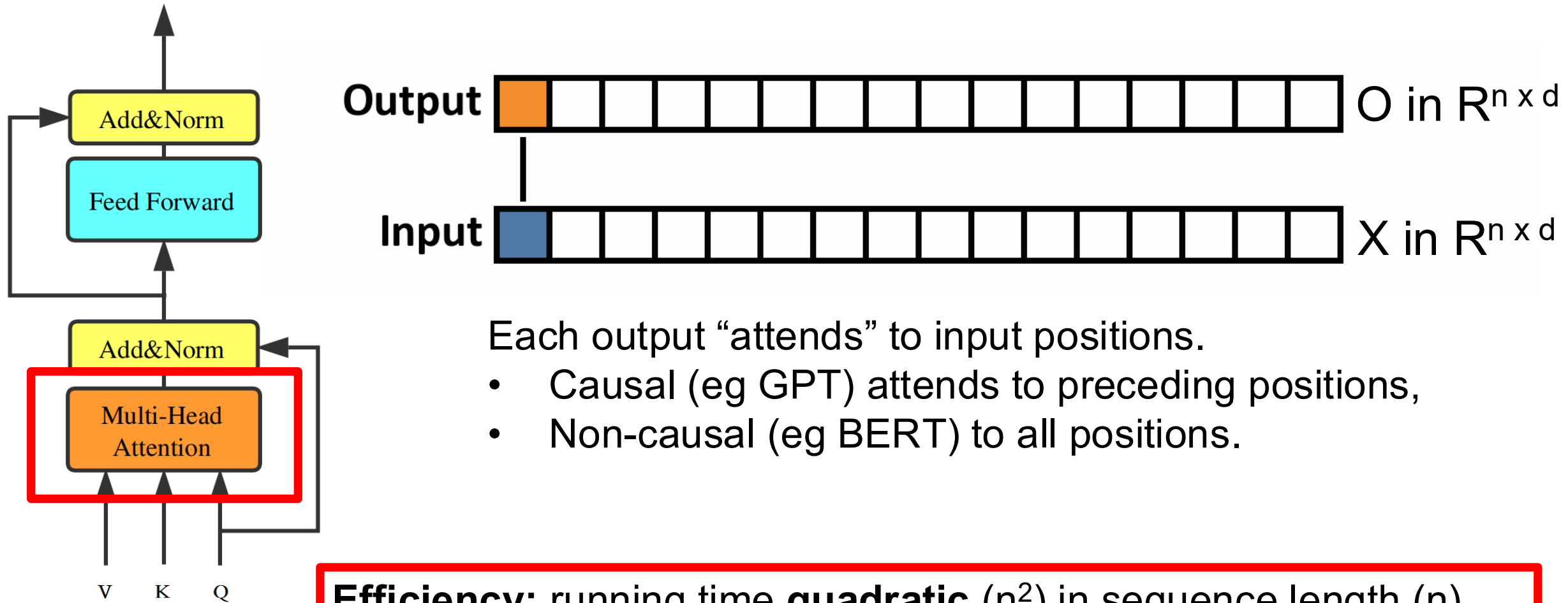
Database people count IO as reads-and-writes from HBM (slow memory).



As accelerators get faster, memory more often the bottleneck.



Attention (Visual Version of Single Head)



Efficiency: running time **quadratic** (n^2) in sequence length (n)



Simple Attention Computation (Single head)

Input : X in $\mathbb{R}^{n \times d}$

Output: O in $\mathbb{R}^{n \times d}$

q, k, v are linear maps of X to matrices:
 Q, K in $\mathbb{R}^{n \times r}$ V in $\mathbb{R}^{n \times d}$
called **Queries**, **Keys**, and **Values**

$Q, K, V = q(X), k(X), v(X)$

$O = \text{SoftMax}(QK^T, \text{dim}=1)V$

$$W_{i,j} = \exp(Q_i \cdot K_j)$$

$$\text{SoftMax}(Q_i; K_1, \dots, K_n) = \frac{W_i}{\sum_{l=1 \dots n} W_{i,l}}$$



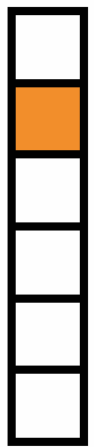
Q



K^T



V



O

Asymmetry: Output O_i depends on just Q_i but **all** of K , each of $K_1, \dots, K_n$ via **normalization**



Flash Attention Main Idea



Tri Dao



Dan Fu

- Minimize IO (HBM to SRAM) – not FLOPs
- Aggressive **fusion**: when you pull in data use it.

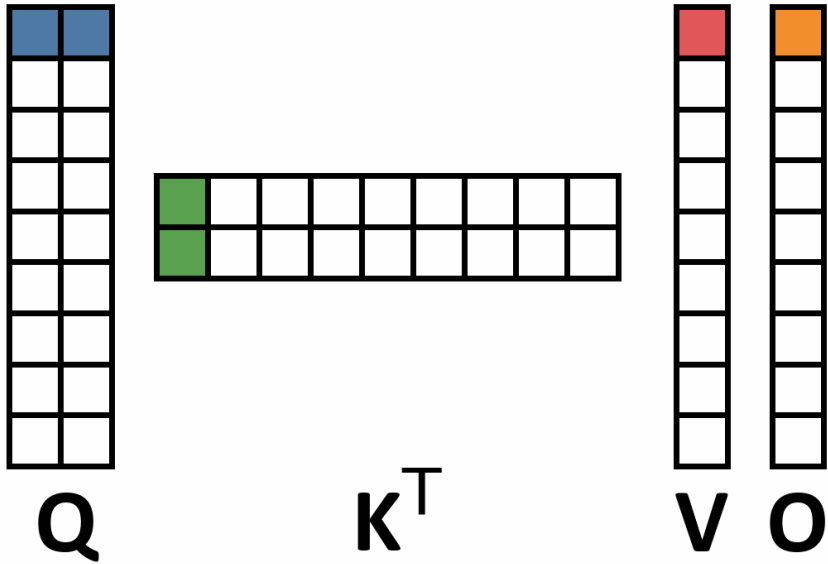
*Two **classical** ideas, I learned from database researchers.*

Up to 72% Utilization—
15% faster BERT on MLPerf 1.1

- Minimize IO (HBM to SRAM) – not FLOPs
- Aggressive **fusion**: when you pull in data use it.



Minimize IO to HBM in Flash Attention



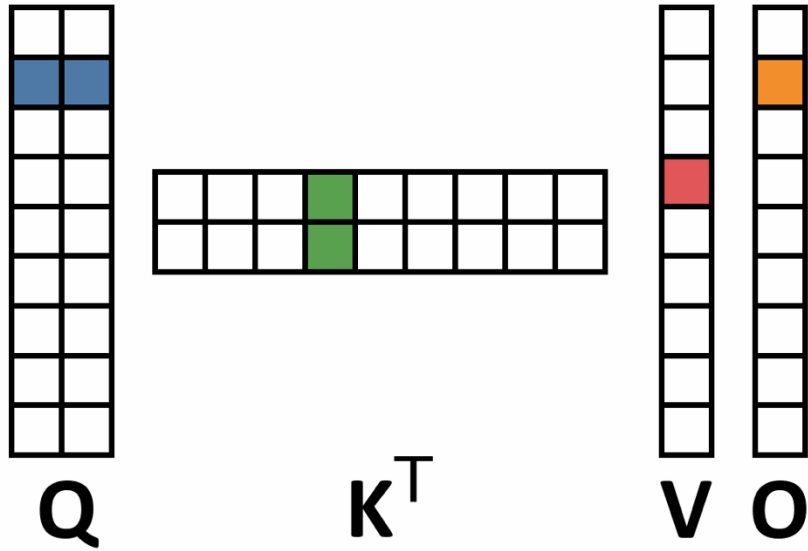
Database people call “*nested loop join*”
 $r(Q) = 9$ is the rows of Q , $|Q|$ is number of tiles.

$$|Q| + r(Q)(|K| + |V|) + |O| = 18 + 9 \times (18 + 9) + 9 = 270 \text{ IO}$$

- Minimize IO (HBM to SRAM) – not FLOPs
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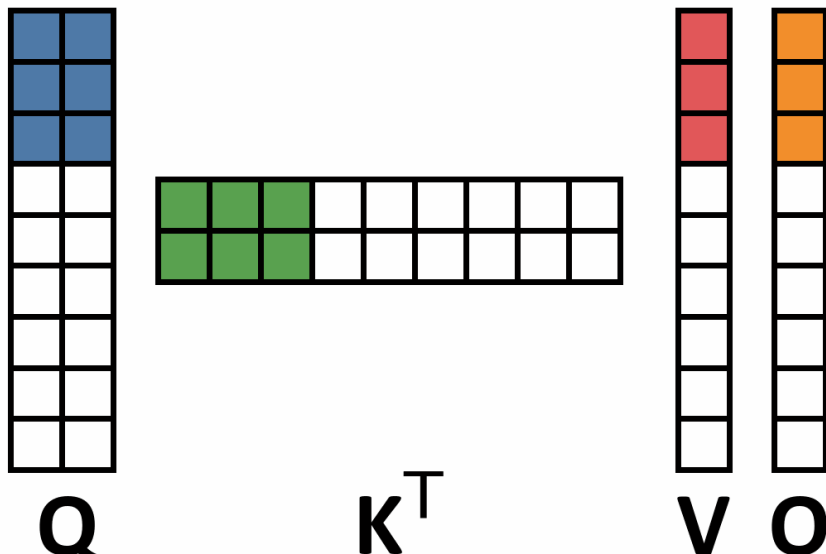
Minimize IO to HBM in Flash Attention



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$$|Q| + r(Q)(|K| + |V|) + |O| = 18 + 9 \times (18 + 9) + 9 = 270 \text{ IO}$$



Database Idea: **Block** Nested Loop Join.

Read 3 blocks at once, so $b(Q) = 9/3 = 3$.

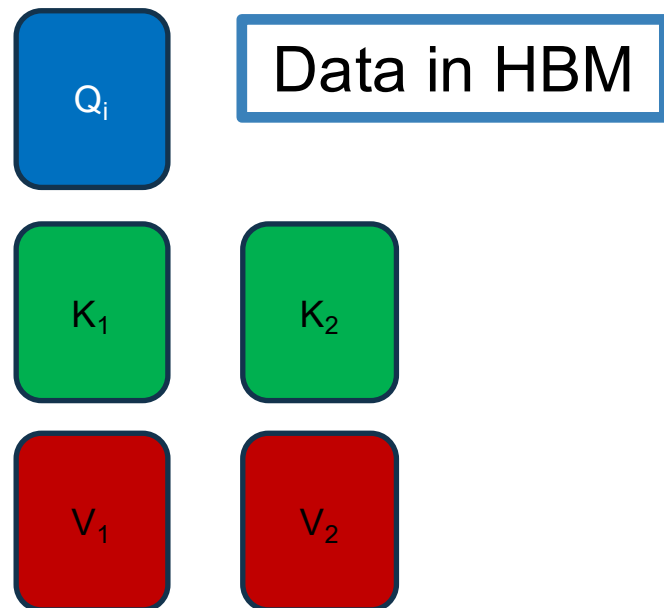
$$|Q| + b(Q)(|K| + |V|) + |O| = 18 + 3 \times (18 + 9) + 9 = 108 \text{ IO}$$

Same FLOPS but ~3x reduction in IO w/ block size 3. Flash Attention A100 uses 8x8 blocks.

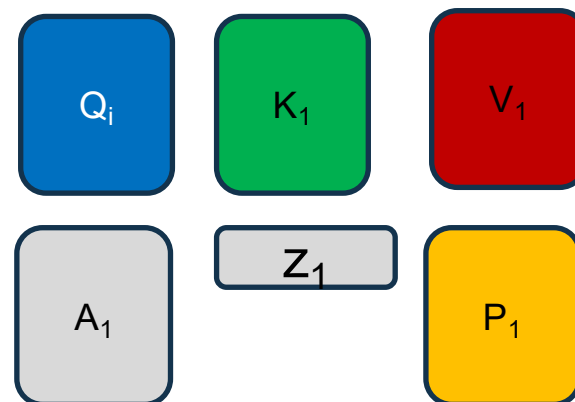
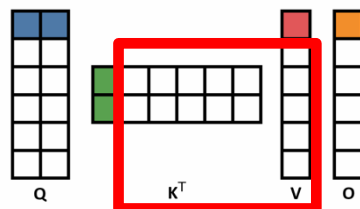


- Minimize IO (HBM to SRAM) – not FLOPs
- Aggressive **fusion**: when you pull in data use it.

IO-Aware Attention



$A = \exp(QK^T)$
 $W = AV$
 $Z = A.\text{sum}(-1)$
 $O = W/Z$



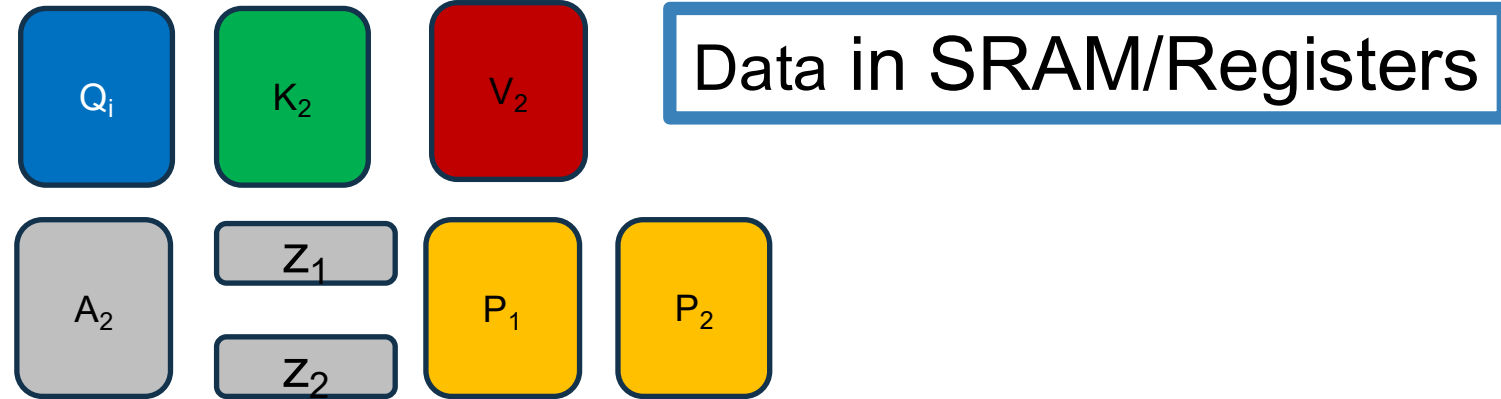
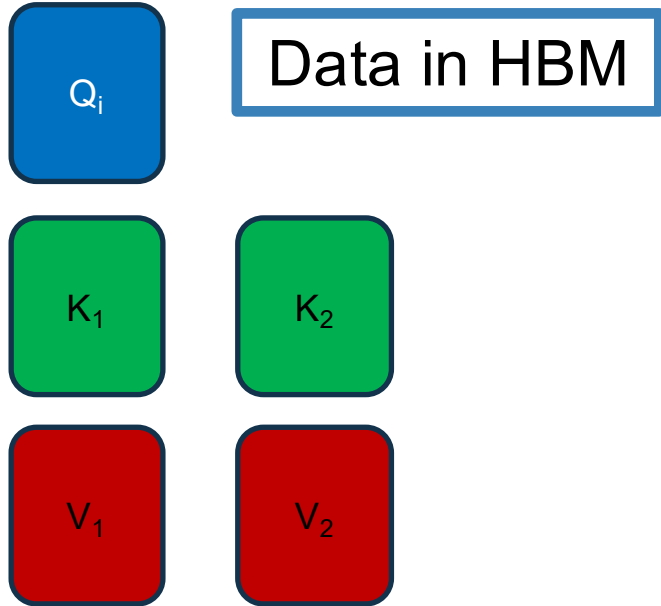
```
Load( $Q_i, K_1, V_1$ )  
 $A_1 = \exp(Q_i K_1^T)$  // attention scores  
 $z_1 = A_1.\text{sum}(-1)$  // normalization  
 $P_1 = A_1 V_1 / z_1$  // compute output.  
Free( $K_1, V_1, A_1$ )
```

Incorrect Normalization! Normalization depends on the rest of K and the output on the rest of V —we haven't seen them!

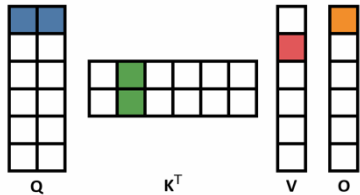


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IO-Aware Attention



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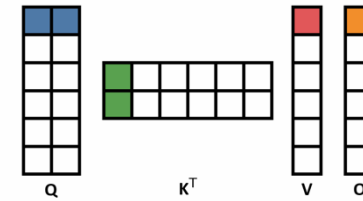
Load(K_2, V_2)
 $A_2 = \exp(Q_i K_2.T)$ // attention scores
 $z_2 = z_1 + A_2.\text{sum}(-1)$ // normalization
 $P_2 = A_2 V_2 / z_2$ // compute output
 $P_2 = P_2 + z_1 / z_2 P_1$ // renormalize P_1 !
 Free(K_2, V_2, A_2, Z_1, P_1).
 After t iterations, we set $O_i = P_t$

NB: To Fix Normalization only need to keep last (P_t, Z_t)

Summary of Block-Aware Attention to FlashAttention



▷ Minimize HBM IO by **blocking**.

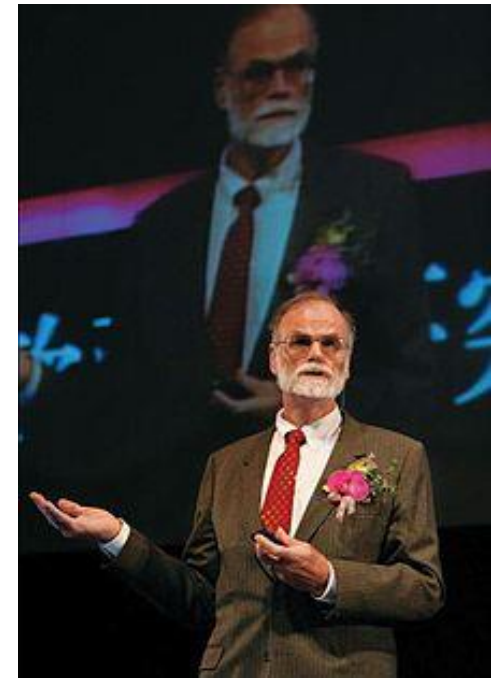


▷ To **fuse**, we **aggregate** like running sum to compute attention exactly.

- In database terms, softmax is an *algebraic aggregation* [Gray07].

```
Load(K2, V2)
A2 = exp(Q2K2.T) // attention
scores
Z2 = z1 + A2.sum(-1) //
normalization
O2 = A2V2/Z2 // compute the
output
O2 = O2 + z1/Z2O1 //
renormalized!
Free(K2, V2, A2, Z1, O1)
```

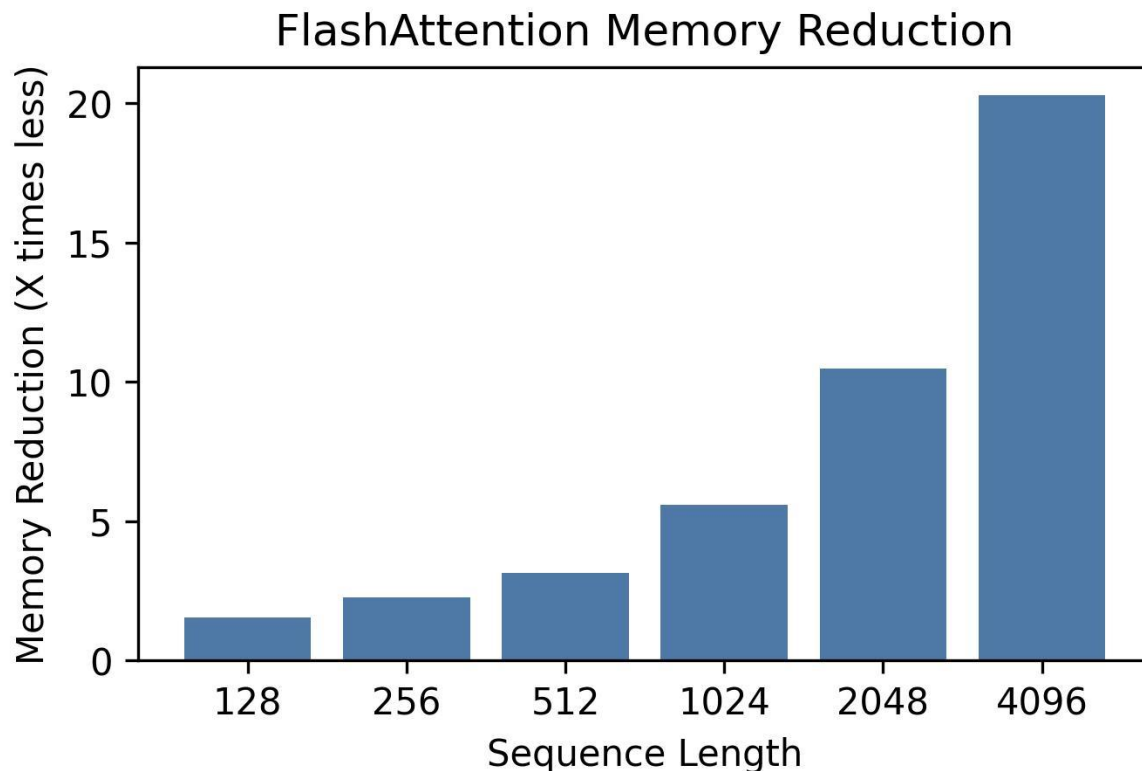
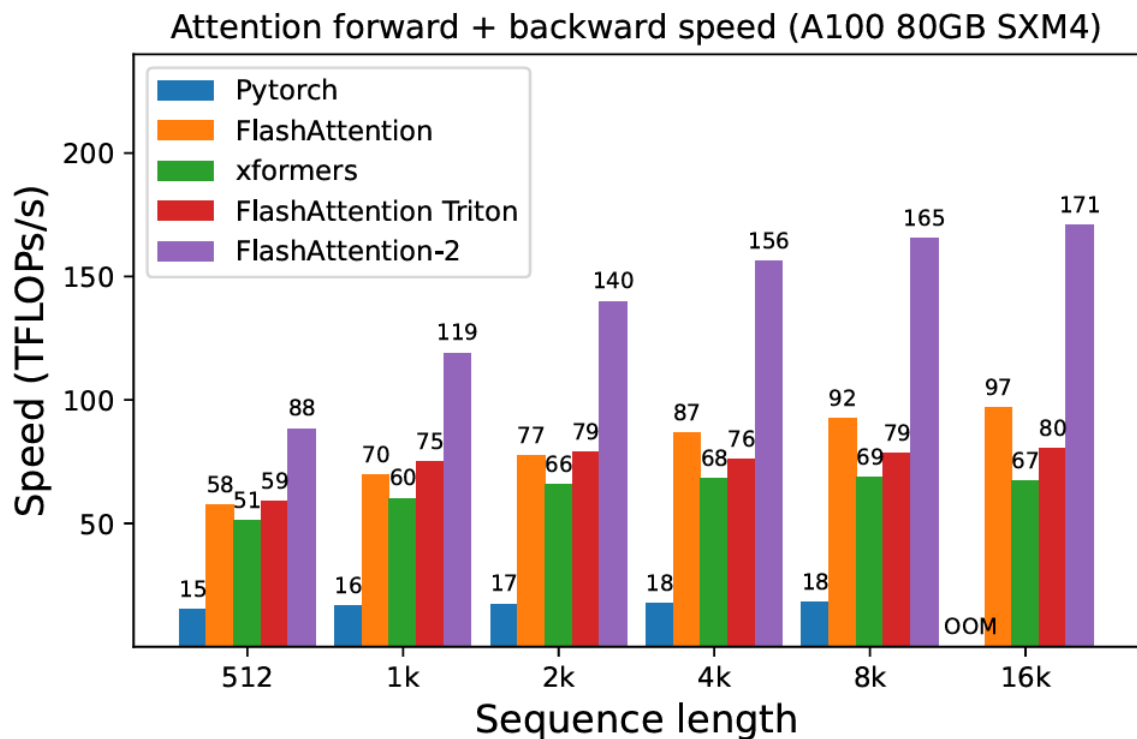
▷ FlashAttention essentially block-nested loop join from classical databases.



Jim Gray, Turing Award 1998.
Personal Hero

There is a whole canon of systems to re-explore for AI!

FlashAttention: 6-10x speedup, 10-20x memory reduction



6-10x speedup — with no approximation

10-20x memory reduction

We're Training AI Twice as Fast This Year as Last > New MLPerf rankings show training times plunging

BY [SAMUEL K. MOORE](#) | 30 JUN 2022 | 5 MIN READ | 

“Performing MLPerf benchmarks is no easy task, and often involves the work of many engineers. But a single graduate student, with some consultation, can do it, too. [Tri Dao](#) was that graduate student.”

ML Perf Winners use it!



PRINCETON
UNIVERSITY

together.ai



Tri Dao



Thanks!

Any questions?